

Homework 3

CS/ECE 374-B Intro to Algorithms & Models of Computation

University of Illinois Urbana-Champaign

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Maximum Points: 100

Due Date: Thursday, February 27, 2025, 11:59 PM

Instructions

1. Answer **ALL** questions.
2. Show all the work.
3. Legibly scribe your solutions and upload them as individual problems on Gradescope.

Suggestions

While writing your answers and/or proofs, think about the following.

1. What is it that you are given? What is it that you are assuming? What is it that you are trying to prove?
2. How exactly does a step follow from the previous step? Assume nothing is apparent, even algebraic simplifications. Your reasoning should be clear and convincing.
3. If you are using a result from the class then state it clearly before using it. You should identify where exactly you need that result, cast your problem into the setup of the result, and then use it.
4. If you are using a result from outside the class then you should prove it before using it

Problems

1. Prove that the following language is not regular:

$$L = \left\{ w : w \in \{0, 1\}^* \text{ such that } |w| = \lceil k\sqrt{k} \rceil, k \in \mathbb{N} \right\},$$

where $\lceil \cdot \rceil$ is the *ceiling* function that returns the smallest integer greater than or equal to a given number. For example: $\lceil 4.2 \rceil = 5$, $\lceil 2.0 \rceil = 2$.

Hint: Considering the difficulty of the problem, we are providing you with a fooling set that works, i.e., $F = \{0^{m^6} \mid m \geq 1\}$. Furthermore, we are providing the following bounds that can help:

Call two strings x, y *adjacent* if $|x| = \lceil k^{1.5} \rceil$ and $|y| = \lceil (k+1)^{1.5} \rceil$ for some $k \in \mathbb{N}$. The difference between the lengths of *adjacent* strings in L is bounded below and above as follows:

$$1.5\sqrt{k} - 1 \leq \lceil (k+1)^{1.5} \rceil - \lceil k^{1.5} \rceil \leq 1.5\sqrt{k} + 3.$$

Given the above hint, you must carefully prove that F is *indeed* a fooling set for L .

Solution: Let F be the set $\{0^{m^6} \mid m \in \mathbb{N}\}$.

We can also write this as $\{0^{\lceil k\sqrt{k} \rceil} \mid k = m^4, m \in \mathbb{N}\}$. Note that each element in F is also an element in L .

Let $x = 0^{m^6}$ and $y = 0^{n^6}$ for some $m < n$.

Let z be the smallest string such that $xz \in L$. By the given bound, $|z| \leq 1.5m^2 + 3$.

Suppose for contradiction $yz \in L$. By the other side of the given bound, we would need $|z| \geq 1.5n^2 - 1$. We can show both of these constraints on z can't be satisfied, since $1 \leq m \leq n - 1$, so

$$1.5m^2 + 3 \leq 1.5(n-1)^2 + 3 = 1.5(n^2 - 2n + 1) + 3 = 1.5n^2 - 1 + (5.5 - 3n) \leq 1.5n^2 - 1$$

So, we know $yz \notin L$, which makes F a valid fooling set. Since F is infinite, L can't be regular.

2. For a string x , let $n_1(x)$ denote the number of *ones* in it, let $n_2(x)$ denote the number of *twos* in it, and let $n_4(x)$ denote the number of *fours* in it.

- (a) Using a *fooling set*, prove that $L_{124} = \{1^k 2^{2k} 4^{4k} : k \in \mathbb{N}\}$ is non-regular.
- (b) Consider the language $\tilde{L}_{124} = \{x \in \{1, 2, 4\}^* : 4n_1(x) = 2n_2(x) = n_4(x)\}$. Using the fact that regular languages are *closed under intersection*, prove that $\tilde{L}_{124}$ is non-regular.

Solution

- (a) Consider set $F = \{1^k 2^{2k} : k \in \mathbb{N}\}$. Trivially it has infinite cardinality. Taking some arbitrary $x, y \in F$, we have that $x = 1^i 2^{2i}$ and $y = 1^j 2^{2j}$ for some $i \neq j$.

We can see that taking $z = 4^{4i}$, $xz \in L_{124}$ while $yz \notin L_{124}$. As such, F is a fooling set for L_{124} of infinite size, and thus L_{124} is non-regular.

- (b) Consider the language $L_{124}^r = 1^* 2^* 4^*$. Clearly it is regular. Note that $L_{124} = \tilde{L}_{124} \cap L_{124}^r$. If $\tilde{L}_{124}$ was regular, by closure of intersection we would have that L_{124} is regular. But that is clearly not the case as we have shown in Part (a)!

We conclude that $\tilde{L}_{124}$ is non-regular. A clever student might note that we could use the same fooling set as in Part (a) to show non-regularity, and partial credit will be given to that solution.

3. Given the alphabet $\Sigma = \{0, 1\}$, for each of the following languages, determine if it is regular. If a language is not regular, write a Context Free Grammar (CFG) that generates it.

(a) $L_1 = \{xwvy \mid x, y, w \in \Sigma^+\}$.

(b) $L_2 = \{0^i 1^{2i} 0^i \mid i \geq 1\}$.

(c) $L_3 = \{0^i 1^j 0^k \mid j > i + k\}$. **Hint:** CFGs are closed under concatenation.

(d) $L_4 = \{0^i 1^j \mid i \text{ is even and } j \text{ is odd}\}$.

Solution

L_1 : Regular.

Explanation: Let's write down some strings that satisfy $xwvy$ and see how we can reallocate each part to x , w , and y .

Example:

$$x = 1, w = 101, y = 010$$

$$xwvy = 1101101010$$

But we can also interpret this string as

$$x = 11011, w = 01, y = 0$$

So the problem becomes: what is the minimal set of w we need to cover all the possibilities? There are only 4 of them.

$$ww_{min} = \{00, 11, 1010, 0101\}$$

Thus it is possible to write a regular expression for this language:

$$(0 + 1)^+(00 + 11 + 0101 + 1010)(0 + 1)^+$$

L_2 : Trick question. Answer “irregular but impossible to write a CFG” for full credit.

This language is irregular but it's impossible to write a CFG for it because it is a context-sensitive language. This question was taken from an MIT Quiz and the original question literally asks students to write a CFG for this – we should have double-checked.

You are not expected to see problems like this during exams.

L_3 : We have

$$S \rightarrow ABC$$

$$A \rightarrow 0A1|\epsilon$$

$$B \rightarrow 1B|1$$

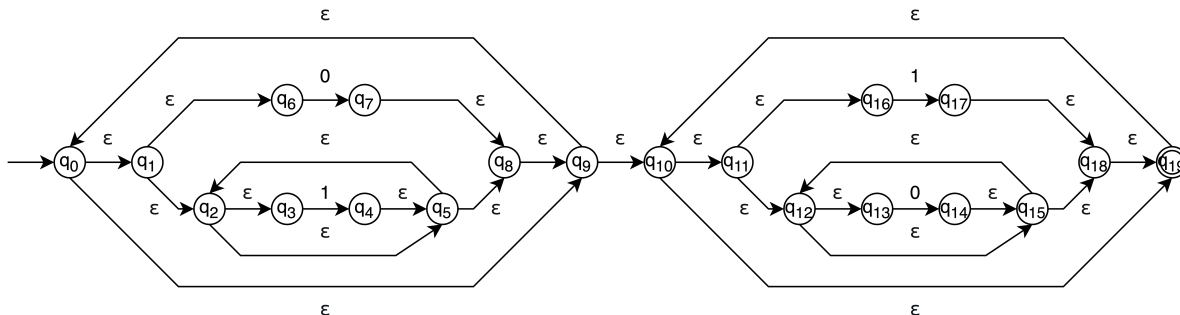
$$C \rightarrow 1C0|\epsilon$$

L_4 : Regular

4. (a) Using *Thompson's algorithm*, convert the regular expression $(0 + 1^*)(1 + 0^*)^*$ into an *equivalent* NFA.
- (b) Using *incremental subset construction algorithm*, convert the NFA you just constructed in Part (a) into an *equivalent* DFA.
- (c) Find the smallest (minimal) DFA that is *equivalent* to your DFA from Part (b) and convert it into a regular expression different from what you started with. What language does this regular expression describe?

Solutions

- (a) A solution is as follows:

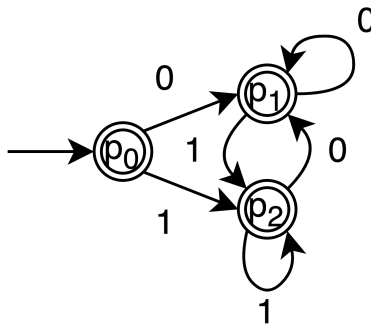


- (b) We start by finding the following ϵ -reach set for q_0 :

$$\epsilon\text{-reach}(q_0) = \{q_0, q_1, q_2, q_3, q_5, q_6, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{15}, q_{16}, q_{18}, q_{19}\}$$

NFA State Set		0	1
p_0	$\epsilon\text{-reach}(q_0)$	$\epsilon\text{-reach}(q_0) \cup \{q_7, q_{14}\}$	$\epsilon\text{-reach}(q_0) \cup \{q_4, q_{17}\}$
p_1	$\epsilon\text{-reach}(q_0) \cup \{q_7, q_{14}\}$	$\epsilon\text{-reach}(q_0) \cup \{q_7, q_{14}\}$	$\epsilon\text{-reach}(q_0) \cup \{q_4, q_{17}\}$
p_2	$\epsilon\text{-reach}(q_0) \cup \{q_4, q_{17}\}$	$\epsilon\text{-reach}(q_0) \cup \{q_7, q_{14}\}$	$\epsilon\text{-reach}(q_0) \cup \{q_4, q_{17}\}$

Note that the only accepting state q_{19} is in $\epsilon\text{-reach}(q_0)$, so all three new states are accepting states. Then, we simply plot out the mapped DFA states with corresponding transitions.



- (c) Since all the states are accepting states, it implies that this DFA is accepting any incoming string, which is just $M = (\{q_0\}, \{0, 1\}, \delta, q_0, \{q_0\})$ with δ being self transitions on consuming either 1 or 0. The language is thus the set of all the strings constructed from 0s and 1s, which can be expressed as $(0 + 1)^*$.

5. For a string x , let $n_0(x)$ denote the number of *zeros* in it, and let $n_1(x)$ denote the number of *ones* in it. Consider the following language.

$$L = \{w : w \in \{0, 1\}^* \text{ and } n_0(w) = n_1(w)\}$$

- (a) Show that L is non-regular.
- (b) Give a context-free grammar that generates L .
- (c) Construct a pushdown automaton that recognizes L .

Solution.

- (a) We will use the concept of fooling sets to prove that the language $L = \{w \in \{0, 1\}^* \mid n_0(w) = n_1(w)\}$ is non-regular.

Let F be the language 0^* .

Let x and y be arbitrary strings in F .

Then $x = 0^i$ and $y = 0^j$ for some non-negative integers $i \neq j$.

Let $z = 1^i$.

Then $xz = 0^i 1^i \in L$.

And $yz = 0^j 1^i \notin L$, because $i \neq j$.

Thus, F is a fooling set for L .

Because F is infinite, L cannot be regular.

- (b) $S \rightarrow \epsilon \mid SS \mid 0S1 \mid 1S0$

- (c) The PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ is defined as follows:

- States: $Q = \{q_0, q_1, q_f\}$
- Input alphabet: $\Sigma = \{0, 1\}$
- Stack alphabet: $\Gamma = \{0, 1, Z_0\}$
- Start state: q_0
- Start stack symbol: Z_0
- Accept states: $F = \{q_f\}$
- Transition function δ :
 - $\delta(q_0, 0, Z_0) = (q_0, 0Z_0)$
 - $\delta(q_0, 1, Z_0) = (q_0, 1Z_0)$
 - $\delta(q_0, 0, 0) = (q_0, 00)$
 - $\delta(q_0, 1, 1) = (q_0, 11)$
 - $\delta(q_0, 0, 1) = (q_0, \epsilon)$
 - $\delta(q_0, 1, 0) = (q_0, \epsilon)$
 - $\delta(q_0, \epsilon, Z_0) = (q_f, Z_0)$

